JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-05 [P1]

KEY SHEET

CHEMISTRY

1)	**A**	2)	**C**	3)	**D**	4)	**C**	5)	**C**	6)	**D**	7)	**C**	8)	**ABC**
9)	**BC**	10)	**ABD**	11)	**ABC**	12)	**A**	13)	**B**	14)	**A**	15)	**A**	16)	**D**
17)	**2**	18)	**6**	19)	**4**	20)	**6**	21)	**7**	22)	**8**	23)	**5**		

PHYSICS

24)	**A**	25)	**B**	26)	**B**	27)	**D**	28)	**C**	29)	**D**	30)	**C**	31)	**ABD**
32)	**BCD**	33)	**AC**	34)	**AC**	35)	**D**	36)	**C**	37)	**B**	38)	**C**	39)	**B**
40)	**8**	41)	**8**	42)	**2**	43)	**4**	44)	**3**	45)	**6**	46)	**5**		

MATHEMATICS

47)	**B**	48)	**C**	49)	**A**	50)	**B**	51)	**A**	52)	**C**	53)	**B**	54)	**BCD**
55)	**BC**	56)	**ABC**	57)	**BCD**	58)	**C**	59)	**B**	60)	**C**	61)	**C**	62)	**D**
63)	**2**	64)	**4**	65)	**9**	66)	**7**	67)	**1**	68)	**5**	69)	**6**		

SOLUTIONS
CHEMISTRY

1. $\Lambda_m = \Lambda_m^0 - b\sqrt{c}$

$260 = \Lambda_m^0 - b\sqrt{0.25}$ ….. 1

$250 = \Lambda_m^0 - b\sqrt{1}$ …….. 2

The above equation 1 and 2

b= 20

$\therefore \Lambda_m^0 = 260 + 20(0.5) = 270$

2. $\boxed{NaCN} + HCl \rightarrow NaCl + \boxed{HCN}$

$P^H = P^{ka} + \log \dfrac{[NaCN]}{[HCN]}$

3. For condensation $\Delta H = -Ve \Rightarrow \Delta S_{sur} = +ve$

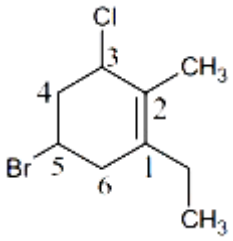
$\Rightarrow \Delta S_{sys} = -ve$

For natural process $\Rightarrow \Delta S_{total} > 0$

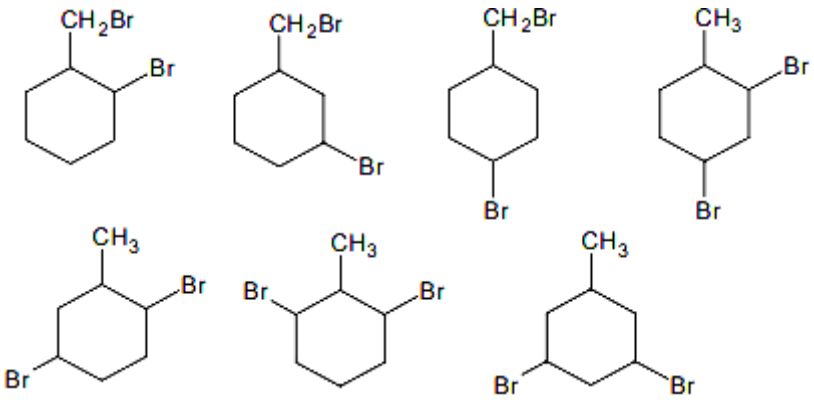
4. $XeF_2 + BF_3 \rightarrow [XeF]^+ [BF_4]^-$

$XeF_6 + CsF \Rightarrow CS^+ [XeF_7]^-$

5.



6.



7. I–resonance; $III - sp^2$ hybridized nitrogen; $IV - sp^2$ hybridized nitrogen with positive charge on nitrogen.

8. $Ce^{4+} + pb^{2+} \rightarrow pb^{4+} + Ce^{+3}$

$E^0 = +0.155V$

$MnO_4^- + Pb^{2+} \rightarrow Pb^{4+} + Mn^{2+}$

$E^0 = +0.055V$

$Ce^{4+} + Mn^{2+} \Rightarrow Ce^{3+} + MnO_4^-$

$E^0 = +0.05484V$

9. $\dfrac{0.02}{V_1} = \dfrac{0.03}{V_2} == \dfrac{0.04}{V_3}$

$V_1 : V_2 : V_3 = 2 : 3 : 4$

$V_1 = \dfrac{2}{9} \times 300 = \dfrac{200}{3} ml$

$V_2 = \dfrac{3}{9} \times 300 = 100 ml$

$V_3 = \frac{4}{9} \times 300 = \frac{400}{3} ml$

10. A. back bonding from O to Si

B. $(CH_3)_3 SiO^{\Theta}$ stabilized by back bonding

C. $P(SiH_3)_3$ is pyramidal

D. ClO_4^- does not polymerize

11. A- Enantimer B- Diastereo C- Identical D- No releationship

12. $P_A = 2P_B$

$P_A + P_B = 2 \Rightarrow P_B = \frac{2}{3}$ *and* $P_A = \frac{4}{3}$

From PV=nRT $n_A = \frac{8}{3}$ & $n_B = \frac{4}{3}$

13.

$A \rightarrow B$

$\frac{8}{3} \quad \frac{4}{3}$

$T = 5_{min}$ $\frac{8}{3} - \frac{4}{3}$ $\frac{4}{3} + \frac{4}{3}$

$\therefore t_{1/2} = 5$

In two half lives 75% completed

$\therefore$ moles of 'A' left $= \frac{1}{4} \times \frac{8}{3} = \frac{2}{3}$

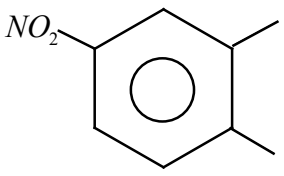
moles of 'B' left $= 2 + \frac{4}{3} = \frac{10}{3}$

$P \times 8.21 = 4 \times 0.821 \times 100$

P=4

14. $\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}} = \frac{\frac{4}{3}}{\frac{2}{3}} = 2$

15.



Br is anti to -OH therefore it will migrate

16. oximes of P are geometrical isomers

Q are metamers

methyl is having less migrating aptitude than ethyl

17. At second end point we have Na_2HPO_4

$pH = \frac{P^{ka_2} + P^{ka_3}}{2} = 10.5$

$\therefore E_{Cell} = 1.3805 + (0.059 \times 10.5) = 2$

18. $10 \times M_1 \times 3 = 25 \times 0.4 \times 5$

$M_1 = 1.66M$

$\Delta T_f = i\,k_f m = 2\times 1.86\times 1.66 = 6.2$

19. $NH_2COONH_4 \rightleftharpoons 2NH_3 + CO_2$

$5L \rightarrow \qquad 4 \qquad 2$

$K_p = 4^2 \times 2 = 32$

$10L \rightarrow \qquad 2(1+p) \qquad 1+p$

$2^2 (1+p)^2 (1+p) = 32$

$(1+p)^3 = 8$

$\Rightarrow p = 1$

Total pressure $3(1+p) = 6$

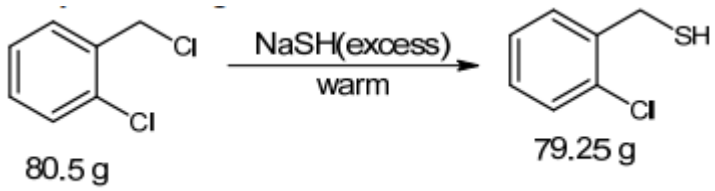
20. $\mu = 2.84D$ for

$[NiCl_4]^{2-}\,[NiCl_2(PPh_3)][NiF_6]^{4-}$

$[Ni(NH_3)_6]^{2+}\,[Ni(cn)_6]^{2+}\,[Ni(H_2O)_6]^{2+}$

21. –COOH group, all OH groups, SH group and H of terminal alkynes react with MeMgBr to give CH_4

22.

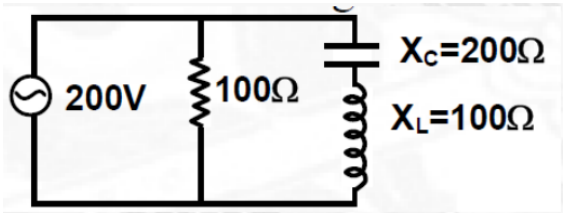


23. a, c, d, e, f are formed optically inactive

PHYSICS

24. $a = K\sqrt{V}; \int V^{\frac{1}{2}} dv = \int_0^s kds; s = \frac{2}{3}\frac{2V_0^{3/2}}{K}$

25.



$i = \sqrt{i_R^2 + i_c^2}$

$\sqrt{2^2 + 2^2} = 2\sqrt{2}$

26. Cut off wavelength $\lambda_c = \frac{hc}{eV}$ where V is the accelerating voltage.

Characteristic wave length $\lambda \propto \frac{1}{(Z-b)^2}$

27. $\Delta W = \int_{(OA)} \bar{F}.\overline{dr} + \int_{(AB)} \bar{F}.\overline{dr}$

$\Delta W = \int_0^4 x^3 dx + \int_0^6 12y dy$

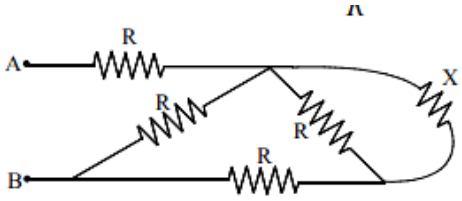
$\Delta W = 152J$

28. Apply Biot Savart's law $B = B_1 + B_2$

B_1 (Due to semi-infinite straight wire) $= \frac{\mu_0 l}{4\pi a}$

B_2 (Due to parabolic portion) $= \frac{\mu_0 l}{8a} B = \frac{\mu_0 l}{4a}\left[\frac{1}{\pi} + \frac{1}{2}\right]$

29.



Let equivalent resistance be x

$$\frac{\left(\frac{xR}{x+R} + R\right)R}{\frac{xR}{x+R} + 2R} + R = x$$

$x^2 - Rx - R^2 = 0, \quad x = \frac{R + \sqrt{5}R}{2}$

30. $\phi = BA\hat{k}.\hat{n}$ $\qquad \phi = \frac{BA}{\sqrt{2}}\cos(wt)$

$\varepsilon = \frac{BA\omega}{\sqrt{2}}\sin(\omega t)$ $\qquad i = \frac{BA\omega}{\sqrt{2}R}\sin(\omega t)$

$\vec{m} = iA\hat{k} = \frac{BA^2\omega}{\sqrt{2}R}\sin(\omega t)\hat{k}$

$\vec{T} = \vec{m} \times \vec{B} = \frac{B^2 A^2 \omega}{\sqrt{2}R} \sin(\omega t)\hat{k}$ $= \frac{B^2 A^2 \omega}{2R}\left[\hat{i}\right] \sin^2(\omega t)$

$T = -\frac{B^2 A^2 \omega}{2R}\left[\sin^2\left(\frac{\pi}{6}\right)\right] = \frac{-\alpha}{4}\hat{i}$

$(I) \rightarrow Q$

$II. \psi = 0$

$II \rightarrow P$

III. $\phi = \frac{BA}{\sqrt{2}} \cos(\omega t)$

$i = \frac{BA\omega}{\sqrt{2}R} \sin(\omega t)$ $\vec{m} = \frac{BA^2 \omega}{\sqrt{2}R} \sin(\omega t)\hat{k}$

$\vec{\tau} = \vec{m} \times \vec{B} - = \frac{B^2 A^2 \omega}{\sqrt{2} \times \sqrt{2}R} \sin \omega t\left(\hat{k} \times \left(\sin \omega t\hat{i} + \cos \omega t\hat{j}\right)\right)$

$\tau = \frac{B^2 A^2 \omega \sin(\omega t)}{2R} \sin(\omega t)\hat{j}$ $= \frac{B^2 A^2 \omega}{2R} \sin(\omega t)\hat{j}$ $= \frac{\alpha}{4}\hat{j}$

$III \rightarrow S$

IV. $\phi = \frac{BA}{\sqrt{2}} \sin(\omega t)$ $i = -\frac{BA\omega}{\sqrt{2}R} \cos(\omega t)$ $vecm = -\frac{BA^2 \omega}{\sqrt{2}R} \cos \omega t\left(\hat{k}\right)$

$\vec{\tau} - = \vec{m} \times \vec{B} - = -\frac{B^2 A^2 \omega}{2R}\left(\hat{k} \times \hat{j}\right) \cos^2(\omega t)$

$\tau = +\frac{B^2 A^2 \omega}{2R}\left(\hat{i}\right).\cos^2\left(\frac{\pi}{6}\right)$ $= +\frac{3}{4}\alpha\hat{i}$ $(IV) \rightarrow R$

31. For the blocks to remain stationary on the inclined surface:

$(m_1 + m_2) g \sin\theta \le \mu_1 m_1 g \cos\theta + \mu_2 m_2 g \cos\theta \Rightarrow \tan\theta \le \frac{\mu_1 m_2 + \mu_2 m_2}{(m_1 + m_2)}$

If N is the force applied by upper block on the lower block, then assuming both are slipping on the incline: $(m_1 + m_2) g \cos\theta - (\mu_1 m_1 g \cos\theta + \mu_2 m_2 g \cos\theta) = (m_1 + m_2) a$

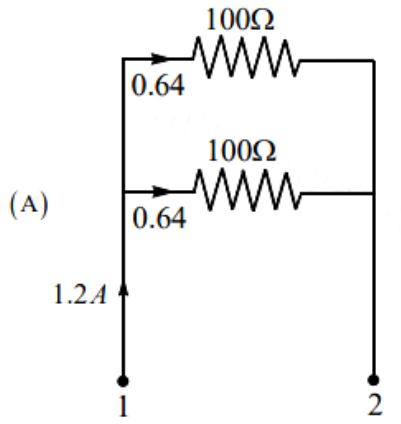
In this situation, using Newton's laws for only lower block:

$(m_1) g \sin\theta + N - (\mu_1 m_1 s \cos\theta) = (m_1) a$

Which shows, $N \neq 0$ Under a situation that $\mu_1 < \mu_2$ with both blocks slipping on incline, the lower block will have more acceleration that upper block causing the blocks to get separated. Thus, the normal force will become zero

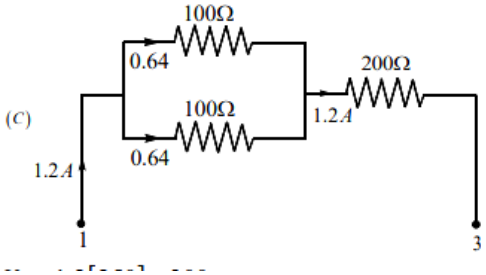
32. Internal force can also change the kinetic energy of system.

33.



$30 units \rightarrow 1.2A$

$1 unit \rightarrow 0.04A$



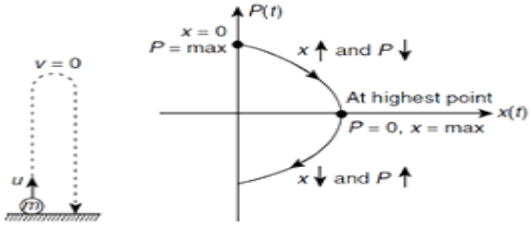
$V_{13} = 1.2[250] = 300 \quad 30 units \rightarrow 300 Volt \quad 1 unit \rightarrow 10 volt$

34. Flux through a Gaussian surface (cylindrical shape) of length *l* and radius 2R

$$\oint \vec{E}.d\vec{s} + 2\oint \vec{E}.d\vec{s} = \frac{\lambda l + (-\sigma)2\pi Rl}{\varepsilon_0}$$

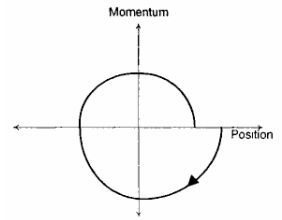
$$\text{Total flux} = \frac{l}{\varepsilon_0}[\lambda - \sigma 2\pi R] = 0 \qquad \sigma = \frac{\lambda}{2\pi R}$$

35. At time *t*=0, the velocity is maximum (maximum momentum) and the displacement is zero. It reaches maximum height when the momentum is zero and falls back with a negative momentum.

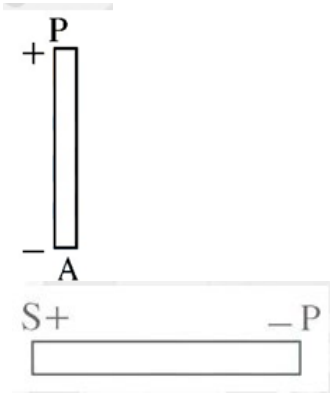


36. $E \alpha p^2$

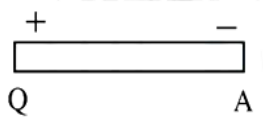
37. It comeback to equilibrium position



38 & 39.



$$Ee = ma \Rightarrow E = \frac{ma}{e} in(SP) \quad Ee = ma \Rightarrow E = \frac{mg}{e} in(PA) \qquad E_{Net} = \frac{mga}{e}$$



$$R_{Net}=\frac{3R}{4}+3R=\frac{15R}{4}=\frac{15Pa}{4S}$$

$$i=\frac{mga}{e\left(\frac{15R}{4}\right)}=\frac{4mgaS}{15ePa}=\frac{4mgS}{15eP}\ \therefore V_{PQ}=\frac{mga}{e}-(2R)\frac{4mga}{15\,\mathrm{Re}}=\frac{7mga}{15e}\quad i_{RS}=\frac{i}{4}=\frac{mgS}{15eP}$$

40. $\frac{l}{2\upsilon}+\frac{\sqrt{4r^2-l^2}}{\upsilon}=80$

41. $F=2mnA\left[(v_0+v)^2-(v_0-v)^2\right]\qquad=2\times10^{-20}\times10^{25}\times1\left[(5+2)^2-(5-2)^2\right]=8\mathrm{N}$

42. Since the boards are perpendicular so the impulses provided by board does not effect the motions Perpendicular to the board. Let T_A be the time internal between successive collisions with board A and from board B is $T_B.T_A=\sqrt{\frac{8a}{g\cos a}};T_B=\sqrt{\frac{8b}{g\cos a}}$

Let the ball collides X ties with wall B form each time it bounces with wall A

$$xT_B=T_A=1x=\sqrt{\frac{a}{b}}\tan a=\sqrt{\frac{4}{1}}\times1=2$$

43. $P_1=1.0atm,V_1=1.0L,T_1=122K$, $\quad P_2=1.0atm,V_2=2.0L,T_2=244K,K=2T_1$

$P_3=2.0atm,V_3=2.0L,T_3=488K,K=4T_1$, $\quad P_4=0.0157atm,V_4=64L,T_4=122K,K=T_1$

For 1 & 2 $\quad Q_1=nC_p\Delta T=\frac{7}{2}nR(T_2-T_1)\ \ \frac{7}{2}\times n\times R(T_1)$

For 2 & 3 $\quad Q_2=nC_p\Delta T=\frac{5}{2}nR(T_3-T_2)\ \ \frac{5}{2}\times n\times R(2T_1)=5nRT_1$

For $3\rightarrow4$ $\quad Q_3=0$, For $4\rightarrow1$ $Q_4=nRT\ln\frac{V_2}{V_1}$

$$=nRT_1\ln\frac{1}{64}=-nRT_1\ln64\quad n=1-\frac{Q-}{Q+}=1-\frac{nRT_1\ln64}{\frac{7}{2nRT_1}+5nRT_1}=1-\frac{6\ln2}{\frac{17}{2}}=1-\frac{12}{17}\ln2$$

44. Total flux coming out of complete closed surface is given as

$$\phi_{hemisphere}+\phi_{cone}=\frac{q}{\varepsilon_0}\ldots\ldots\ldots..\ (1)$$

Through the hemisphere, half of flux originated by q will come out, given as

$\phi_{hemisphere}=\frac{q}{2\varepsilon_0}$.Thus equation-(1), flux coming out of conical surface is given as $\phi_{come}=\frac{q}{2\varepsilon_0}$

$$\Rightarrow\frac{nq}{6\varepsilon_0}=\frac{q}{2\varepsilon_0}\ \Rightarrow n=3$$

45. $\tau_{net}=\frac{-3R}{8}\frac{1}{3}\pi\frac{3R^2}{4}\frac{3R}{2}\times8\rho_0+\frac{4}{3}\pi R^3\rho\omega gR$

$\tau_{net}=\frac{-3}{2}\pi R^4\times10^4+\frac{4}{3}\pi R^4 10^4\qquad\tau_{net}=\frac{1}{6}\pi R^4\times10^4\ \tau_{net}=21.6\pi\times10^5\,Nm$

46. All the elements of circuit are in parallel arrangement

$\frac{1}{R_{eq}}=\frac{1}{40}\times\frac{1}{40}\times\frac{1}{40}\times\frac{1}{40}\times\frac{1}{20}\times\frac{1}{20}\ =\frac{4}{40}\times\frac{2}{20}\ =5\Omega\qquad Power=\frac{V^2}{R}=20W$

47. we know that $\cos\theta = \dfrac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$ *and* $\sin\theta = \dfrac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

$\therefore \tan\dfrac{\theta}{2}, \tan\dfrac{\phi}{2}$ (are unequal) are the roots of

$(x-a)\dfrac{1-t^2}{1+t^2} + y\dfrac{2t}{1+t^2} = a \Rightarrow xt^2 - 2yt + 2a - x = 0$

$\therefore \tan\dfrac{\phi}{2}\tan\dfrac{\theta}{2} = \dfrac{2a-x}{x} \quad .....(i)$

and $\tan\dfrac{\theta}{2} + \tan\dfrac{\phi}{2} = \dfrac{2y}{x} \quad .....(ii)$

Now $\left(\tan\dfrac{\theta}{2} - \tan\dfrac{\phi}{2}\right)^2 = \left(\tan\dfrac{\theta}{2} + \tan\dfrac{\phi}{2}\right)^2 - 4\tan\dfrac{\theta}{2}\tan\dfrac{\phi}{2}.$ *But* $\tan\dfrac{\theta}{2} - \tan\dfrac{\phi}{2} = 2e$

$\therefore (2e)^2 = \left(\dfrac{2y}{x}\right)^2 - 4\left(\dfrac{2a-x}{x}\right) \Rightarrow 4e^2 = \dfrac{4y^2}{x^2} - 4\left(\dfrac{2a-x}{x}\right)$

$\Rightarrow e^2 = \dfrac{y^2}{x^2} - \dfrac{2a-x}{x}, i.e, y^2 = 2ax - (1-e^2)x^2$

48. Consider the first relation given in the problem carefully. We will apply this relation to itself, so that

$f(x,y) = f(2x+2y, 2y-2x)$

$= f(a,b) (where\, a = 2x+2y, b = 2y-2x)$

$= f(2a+2b, 2b-2a)$

$= f(8y, -8x)$

Again,

$f(x,y) = f(8y,-8x)$

$= f(c,d)\,(where\, c = 8y, d = -8x)$

$= f(8d, -8c)$

$= f(-64x, -64y)$

$= f(4096x, 4096y)$

$= f(2^{12}x, 2^{12}y)$

Now, if we put y = 0 and substitute 2^x in place of x, we have

$f(2^x, 0) = f(2^{12+x}, 0) \Rightarrow g(x) = g(x+12)$

Thus, $g(x)$ is periodic with period 12. The correct option is (c).

49. Tangent to the ellipse at $P(a\cos\alpha, a\sin\alpha)$ is $\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$

Tangent to the circle at $Q(a\cos\alpha, a\sin\alpha)$ is $\cos ax + \sin ay = a$

Now angle between tangents is θ,

Then $\tan\theta = \left|\dfrac{-\frac{b}{a}\cot\alpha - (-\cot\alpha)}{1 + \left(-\frac{b}{a}\cot\alpha\right)(-\cot\alpha)}\right|$

$$= \left|\frac{\cot\alpha\left(1-\frac{b}{a}\right)}{1+\frac{b}{a}\cot^2\alpha}\right| = \left|\frac{a-b}{a\tan\alpha + b\cot\alpha}\right| = \left|\frac{a-b}{\left(\sqrt{a\tan\alpha} - \sqrt{b\cot\alpha}\right)^2 + 2\sqrt{ab}}\right|$$

Now the greatest value of the above expression is $\left|\dfrac{a-b}{2\sqrt{ab}}\right|$ when

$\sqrt{a\tan\alpha} = \sqrt{b\cot\alpha}$

$\Rightarrow \theta_{\max imum} = \tan^{-1}\left(\dfrac{a-b}{2\sqrt{ab}}\right)$

50. (b) Let $n_1 \, and \, n_2$ be the vectors normal to the faces OAB and ABC.

Then

$$n_1 = OA \times OB = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = 5\hat{i} - \hat{j} - 3\hat{j}$$

$$and \; n_2 = AB \times AC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ -2 & -1 & 1 \end{vmatrix} = \hat{i} - 5\hat{j} - 3\hat{k}$$

If θ is the angle between the faces OAB and ABC, then

$\cos\theta = \dfrac{n_1 . n_2}{|n_1||n_2|}$

$\Rightarrow \cos\theta = \dfrac{5+5+9}{\sqrt{25+1+9}\sqrt{1+25+9}} = \dfrac{19}{35} \Rightarrow \theta = \cos^{-1}\left(\dfrac{19}{35}\right)$

51.

$f(x) = f'(x) + f''(x) + f'''(x) + ...\infty$

$f'(x) = f''(x) + f'''(x) + f''''(x).....\infty$

$\therefore 2f'(x) = f'(x) + f''(x) + f'''(x) + ...$

$\Rightarrow 2f'(x) = f(x) \Rightarrow \dfrac{f'(x)}{f(x)} = \dfrac{1}{2}$

On integrating w.r.t x; we get $lnf(x)=\frac{1}{2}x+c$

$x=0; f(0)=1\Rightarrow c=0$

Hence $lnf(x)=\frac{x}{2}\Rightarrow f(x)=e^{\frac{x}{2}}$

52. $G(0,t)=0\ for\ t\geq 0\ so\ g(0)=\int_0^1 f(t).\,0\ dt=0$

$G(1,t)=t.(1-1)=0\ for\ t<1.\ Hence\ g(1)=\int_0^1 f(t).0\,dt,$

$$Also\ \ g(x)=\int_0^x f(t)t(x-1)dt+\int_x^1 f(t)x(t-1)dt=(x-1)\int_0^x tf(t)dt+x\int_x^1 f(t)(t-1)dt$$

$$Hence,\ \ g'(x)=(x-1)xf(x)+\int_0^x tf(t)dt+\int_x^1 f(t)(t-1)dt-xf(x)(x-1)$$

$Thus\ g''(x)=f(x)$

53. As the roots are of opposite signs, product of roots < 0.

$a^2+b^2-1<0 \ \ \Rightarrow\ a^2+b^2<1\ \ \Rightarrow |a+ib|<1$

So, the point $a+ib$ lies inside a circle of centre (0,0) and radius 1.

54. $\frac{5^{a_{n+1}}}{5^{a_n}}=\frac{3n+5}{3n+2}$

$$\frac{5^{a_2}}{5^{a_1}}\times\frac{5^{a_3}}{5^{a_2}}\times.....\times\frac{5^{a_n}}{5^{a_{n-1}}}=\frac{8}{5}\times\frac{11}{8}\times\frac{14}{11}\times.....\frac{3n+2}{3n-1}$$

$$\frac{5^{a_n}}{5^{a_1}}=\frac{3n+2}{5}\quad\Rightarrow 5^{a_n}=3n+2\Rightarrow a_n=\log_5(3n+2)$$

$$[a_n]=3\qquad\Rightarrow\log_5(3n+2)\in[3,4)$$

$$\Rightarrow n\in\{41,42,.....207\}$$

$$[a_n]=4\Rightarrow\frac{623}{3}\leq n<\frac{3123}{3}$$

$$\Rightarrow\in\{208,209,.....,1041\}$$

55. A) Order of the differential equation is 2

B) $\frac{xdy-ydx}{\sqrt{x^2+y^2}}=dx\Rightarrow\frac{\frac{xdy-ydx}{x^2}}{\sqrt{\left(1+\frac{y^2}{x^2}\right)}}=dx$

$$\therefore\ \ln\left|\frac{y}{x}+\sqrt{1+\frac{y^2}{x^2}}\right|=\ln|Cx|;\ \frac{y}{x}+\frac{\sqrt{x^2+y^2}}{x}=Cx\ \ i.e\ \ y+\sqrt{x^2+y^2}=Cx^2$$

C) $y=e^x(A\cos x+B\sin x)$

$$\frac{dy}{dx}=e^x\left(A\cos x+B\sin x\right)+e^x\left(-A\sin x+B\cos x\right)=y+e^x\left(-A\sin x+B\cos x\right)$$

$$\frac{d^2y}{dx^2}=\frac{dy}{dx}+e^x\left(-A\sin x+B\cos x\right)-y$$

$$\frac{dy}{dx}+\left(\frac{dy}{dx}-y\right)-y=2\left(\frac{dy}{dx}-y\right)$$

D) $\left(1+y^2\right)\frac{dy}{dx}+x=2e^{\tan^{-1}y}\Rightarrow\frac{dx}{dy}+\frac{1}{1+y^2}x=\frac{e^{\tan^{-1}y}}{1+y^2}$

$$IF=e^{\int\frac{1}{1+y^2}dy}=e^{\tan^{-1}y}\Rightarrow xe^{\tan^{-1}y}=2\int e^{\tan^{-1}y}.\frac{e^{\tan^{-1}y}}{1+y^2}dy\Rightarrow xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+k$$

56. $\left(x-6\right)^4+\left(x-8\right)^4=16.\ Let\ x-6=y+1\Rightarrow x-8=y-1$

$Now,\left(y+1\right)^4+\left(y-1\right)^4=16\Rightarrow y^4+6y^2-7=0\Rightarrow y^2+7=0\ and\ y^2-1=0$

$y=\pm\sqrt{7}i\ or\ y=\pm1\Rightarrow x=7\pm1 or\ x=7\pm\sqrt{7}i\Rightarrow a=7,b=1,c=7,d=7$

57. $\left({}^nC_1\ {}^nC_2...{}^nC_{n-1}\right)^{\frac{1}{n-1}}<\frac{{}^nC_1+{}^nC_2+...+{}^nC_{n-1}}{n-1}\Rightarrow P<\left(\frac{2^n-2}{n-1}\right)^{n-1}$

$$Also,\left({}^nC_0\ {}^nC_1\ {}^nC_2...{}^nC_{n-1}\ {}^nC_n\right)^{\frac{1}{n+1}}<\frac{{}^nC_0+{}^nC_1+{}^nC_2+...+{}^nC_n}{n+1}\Rightarrow P<\left(\frac{2^n}{n+1}\right)^{n+1}$$

58. $a=20-6\pi,b=10\pi-30,c=\sin^{-1}\sin\left(4\pi-10\right)=10-3\pi$

$So,a+b+c=\pi$

$\sin^{-1}\sin x\geq\left|x-\pi\right|\Rightarrow x\in\left[\frac{\pi}{2},\pi\right]$

So, largest integer x = 3.

59. $5\sec^{-1}x+10\sin^{-1}y=10\pi$

$\Rightarrow\sec^{-1}x=\pi\ and\ \sin^{-1}y=\frac{\pi}{2}$

$\Rightarrow x=-1,y=1$

$So,\tan^{-1}\left(-1\right)+\cos^{-1}\left(0\right)=-\frac{\pi}{4}+\frac{\pi}{2}=\frac{\pi}{4}$

60. For C_n circle let centre by $\left(\alpha_n,1\right)$ and it touches line $L_n\ and\ L_{n+1}$

$i.e\frac{x}{x_n}+\frac{y}{4}=1,\frac{x}{x_{n+1}}+\frac{y}{4}=1$

$$\Rightarrow \frac{4\alpha_n + x_n - 4x_n}{\sqrt{16 + x_n^2}} = 1 \qquad .....(i)$$

$$\Rightarrow \frac{4\alpha_n + x_{n+1} - 4x_{n+1}}{\sqrt{16 + x_{n+1}^2}} = -1 \quad .....(ii)$$

$$\Rightarrow 3\left(x_{n+1} - x_n\right) = \sqrt{16 + x_n^2} + \sqrt{16 + x_{n+1}^2}$$

$$Let\ x_n = 4\cot\theta_n 0 < \theta_n < \frac{\pi}{2} \qquad \Rightarrow \quad \tan\left(\frac{\theta_{n+1}}{2}\right) = \frac{1}{2}\tan\left(\frac{\theta_n}{2}\right)$$

$$As\ \ x_1 = 3 \ \Rightarrow \tan\frac{\theta_1}{2} = \frac{1}{2} \qquad \Rightarrow \tan\left(\frac{\theta_n}{2}\right) = \frac{1}{2^n} \Rightarrow x_n = \left(\frac{2^{2^n} - 1}{2^{n-1}}\right) \ \ x_2 = \frac{15}{2}$$

Putting is (i), we got $\alpha_2 = \frac{31}{4}$

$$\lim_{n\to\infty}\frac{x_n}{2^n} = \lim_{n\to\infty}\frac{2^{2^n-1}}{2^{n-1}2^n} = 2$$

61. For C_n circle let centre by $(\alpha_n, 1)$ and it touches line $L_n\ and\ L_{n+1}$

$$i.e\frac{x}{x_n} + \frac{y}{4} = 1, \frac{x}{x_{n+1}} + \frac{y}{4} = 1$$

$$\Rightarrow \frac{4\alpha_n + x_n - 4x_n}{\sqrt{16 + x_n^2}} = 1 \qquad .....(i)$$

$$\Rightarrow \frac{4\alpha_n + x_{n+1} - 4x_{n+1}}{\sqrt{16 + x_{n+1}^2}} = -1 \quad .....(ii)$$

$$\Rightarrow 3\left(x_{n+1} - x_n\right) = \sqrt{16 + x_n^2} + \sqrt{16 + x_{n+1}^2}$$

$$Let\ x_n = 4\cot\theta_n 0 < \theta_n < \frac{\pi}{2} \qquad \Rightarrow \quad \tan\left(\frac{\theta_{n+1}}{2}\right) = \frac{1}{2}\tan\left(\frac{\theta_n}{2}\right)$$

$$As\ \ x_1 = 3 \ \Rightarrow \tan\frac{\theta_1}{2} = \frac{1}{2} \qquad \Rightarrow \tan\left(\frac{\theta_n}{2}\right) = \frac{1}{2^n} \Rightarrow x_n = \left(\frac{2^{2^n} - 1}{2^{n-1}}\right) \ \ x_2 = \frac{15}{2}$$

Putting is (i), we got $\alpha_2 = \frac{31}{4}$

$$\lim_{n\to\infty}\frac{x_n}{2^n} = \lim_{n\to\infty}\frac{2^{2^n-1}}{2^{n-1}2^n} = 2$$

62. For C_n circle let centre by $(\alpha_n, 1)$ and it touches line $L_n\ and\ L_{n+1}$

$$i.e\frac{x}{x_n} + \frac{y}{4} = 1, \frac{x}{x_{n+1}} + \frac{y}{4} = 1$$

$$\Rightarrow \frac{4\alpha_n + x_n - 4x_n}{\sqrt{16+x_n^2}} = 1 \qquad .....(i)$$

$$\Rightarrow \frac{4\alpha_n + x_{n+1} - 4x_{n+1}}{\sqrt{16+x_{n+1}^2}} = -1 \quad .....(ii)$$

$$\Rightarrow 3\left(x_{n+1} - x_n\right) = \sqrt{16+x_n^2} + \sqrt{16+x_{n+1}^2}$$

$$Let\ x_n = 4\cot\theta_n, 0 < \theta_n < \frac{\pi}{2} \qquad \Rightarrow \quad \tan\left(\frac{\theta_{n+1}}{2}\right) = \frac{1}{2}\tan\left(\frac{\theta_n}{2}\right)$$

$$As\ \ x_1 = 3 \ \Rightarrow \tan\frac{\theta_1}{2} = \frac{1}{2} \qquad \Rightarrow \tan\left(\frac{\theta_n}{2}\right) = \frac{1}{2^n} \Rightarrow x_n = \left(\frac{2^{2n}-1}{2^{n-1}}\right) \Rightarrow x_2 = \frac{15}{2}$$

Putting is (i), we got $\alpha_2 = \frac{31}{4}$

$$\lim_{n\to\infty}\frac{x_n}{2^n} = \lim_{n\to\infty}\frac{2^{2^n-1}}{2^{n-1}2^n} = 2$$

63. We observe that

$$\left(1-x\right)\left(1+x\right)\left(1+x^2\right)\left(1+x^4\right)....\infty = \lim_{n\to\infty}\left(1-x^{2^n}\right) = 1 \ \ \left(\sin ce\left|x\right| < 1\right)$$

Thus, $\left(1+x\right)\left(1+x^2\right)\left(1+x^4\right)....\infty = \frac{1}{1-x}$

We now take the log of both sides and differentiate:

$$\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + ...\infty = \frac{1}{1-x}$$

Thus, $f\left(x\right) = \frac{1}{1-x}$. This is a hyperbola, which touches the given circle at (2, -1) and (0,1) (verify).

$$\Rightarrow \sum_{r=1}^{i}\left[\alpha_r\right] = 0 + 2 = 2$$

64. Let θ_i be the event that is the upper left corner of required square

$$\Rightarrow P\left(\theta_1 \cup \theta_2 \cup \theta_3 \cup \theta_4\right) = \frac{95}{512} \Rightarrow P\left(\theta_1^c \cap \theta_2^c \cap \theta_3^c \cap \theta_4^c\right) = \frac{417}{512}$$

65. Suppose there are T_n ways to cover n steps

$$T_n = T_{n-1} + T_{n-2} + T_{n-3}, T_1 = 1, T_2 = 2$$

66.

$$l = \lim_{n\to\infty} 18\left(\frac{3}{3^2} + \frac{3^2}{3^4} + \frac{3^3}{3^6} + ...\right) = 18 \times \frac{1}{3\left(1-\frac{1}{3}\right)} = 9$$

$$m = \lim_{n\to\infty} 12\left(\frac{2}{2^2} + \frac{2^2}{2^4} + ...\right) = 12 \times \frac{1}{2\left(1-\frac{1}{2}\right)} = 12 \Rightarrow A_n = \frac{9+12}{3} = 7$$

67. $A\left(at_1^2, 2at_1\right), B\left(at_2^2, 2at_2\right), C\left(at_3^2, 2at_3\right), D\left(at_4^2, 2at_4\right)$

$\angle ABC = 90^0 \Rightarrow \dfrac{2}{t_1 + t_2} \times \dfrac{2}{t_2 + t_3} = -1 \Rightarrow \left(t_1 + t_2\right)\left(t_2 + t_3\right) = -4$

$\angle ABC = 90^0 \Rightarrow \dfrac{2}{t_1 + t_4} \times \dfrac{2}{t_3 + t_4} = -1 \Rightarrow \left(t_1 + t_4\right)\left(t_3 + t_4\right) = -4$

t_2, t_4 are roots of

$\left(t + t_1\right)\left(t + t_3\right) + 4 = 0 \Rightarrow t^2 + \left(t_1 + t_3\right)t + t_1 t_3 + 4 = 0 \Rightarrow t_2 + t_4 = -\left(t_1 + t_3\right)$

68. Use the concept, $\left[t\right] \le t < \left[t\right] + 1,\ t \in R$

$So, 3x - 7 \le x^2 - 3x + 2 < \left(3x - 7\right) + 1 \, and \, 3x \in I$

Solving, $x \in \left\{\dfrac{7}{3}, \dfrac{8}{3}, \dfrac{9}{3}, \dfrac{10}{3}, \dfrac{11}{3}\right\}$

69. $\begin{vmatrix} 1 & -\sin\theta & -\cos\theta \\ -\cos\theta & 1 & -1 \\ -\sin\theta & -1 & 1 \end{vmatrix} = 0$

Applying $C_3 \to C_3 + C_2$, then

$\begin{vmatrix} 1 & ... & -\sin\theta & -\sin\theta - \cos\theta \\ -\cos\theta & & 1 & 0 \\ -\sin\theta & & -1 & 0 \end{vmatrix} = 0$

Expanding along C_3, then

$\left(-\sin\theta - \cos\theta\right)\left(\cos\theta + \sin\theta\right) = 0$

$\Rightarrow \quad \left(\sin\theta + \cos\theta\right)^2 = 0$

$\Rightarrow \quad \sin\theta + \cos\theta = 0$

$\Rightarrow \quad \sin\theta = -\cos\theta$

$\therefore \quad \tan\theta = -1$

$\Rightarrow \quad \theta = \dfrac{3\pi}{4} \qquad \left[\because \theta \in \left[0, \pi\right]\right]$

Hence, $\dfrac{8\theta}{\pi} = 6$